Lesson 9.3 Guided Notes

Session 1 · Equations from Tables

I can write an equation that matches a table of values. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
equation / ecuación	A math sentence with an _____ sign.	y = 5x
constant / constante	A number that _____ changes.	the 5 in y = 5x
rate / tasa	The amount for _____ one unit.	$3 for each pound

Key idea

If y ÷ x is the _____ in every row, the equation is y = (that number) × x.

If y − x is the same in every row, the equation is y = x _____ (that number).

Check the equation with _____ rows.

Worked example

Write an equation for the pairs (1, 6), (2, 12) and (5, 30).

6 ÷ 1 = _____, 12 ÷ 2 = 6, 30 ÷ 5 = 6

The rate is _____ in every row.

Answer: y = _____. Check: 6 × 5 = 30.

You try

1. Write an equation for each table.
a. x: 2, 4, 6 y: 10, 20, 30 y = _____
b. x: 1, 2, 3 y: 9, 10, 11 y = _____

_____

2. A table shows x: 3, 5, 8 and y: 24, 40, 64. Write the equation. Then find y when x = 12.

_____

Sum it up: To write an equation from a table, I ___, and I check it by ___.

_____

Session 2 · Equations from Situations

I can write an equation from a described situation and identify its variables. · 6.EE.C.9

Word / Spanish	Meaning — complete the blank	Example
unit rate / tasa unitaria	The amount for exactly _____ unit.	$4 for each hour
multiplicative rule / regla multiplicativa	_____ the input by the same number.	y = 6x
additive rule / regla aditiva	_____ the same number to the input.	y = x + 6
per / por	Means “for _____ one.”	25 miles per gallon

Key idea

A rate (“each,” “per”) _____ the variable.

A one-time fee or a starting amount is _____ once.

Worked example

A pet sitter earns $9 each day. Write an equation for the dollars e earned in d days.

Define the letters: d = days, e = _____.

The rate is $9 for each _____, so e = _____.

Answer: Test d = 5: e = 9 × 5 = _____ dollars.

You try

1. Write an equation for each situation.
a. Each box holds 8 pens. Pens p in b boxes: p = _____
b. A gym costs $10 to join plus $20 a month. Cost c for m months: c = _____

_____

2. Devin earns $13 for each lawn he mows. Write an equation for his pay p for n lawns. Find his pay for 6 lawns.

_____

Sum it up: In my equation, the rate goes ___, and a one-time fee goes ___.

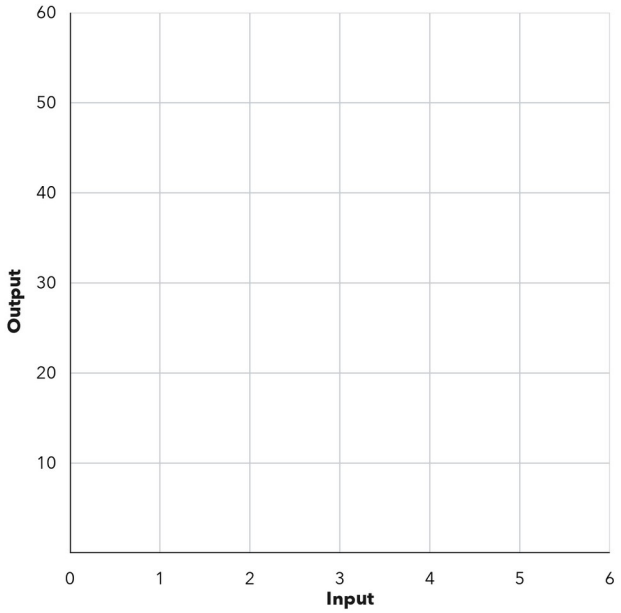
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Session 1 · Model, reason, and explain

Equations from Tables · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare tables A: (0,0),(2,10),(5,25) and B: (0,5),(2,7),(5,10). Write a multiplicative rule for A and an additive rule for B. Check all rows.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: confuses constant difference with constant ratio or divides by zero.. Explain what changes and what stays true.

_____

_____

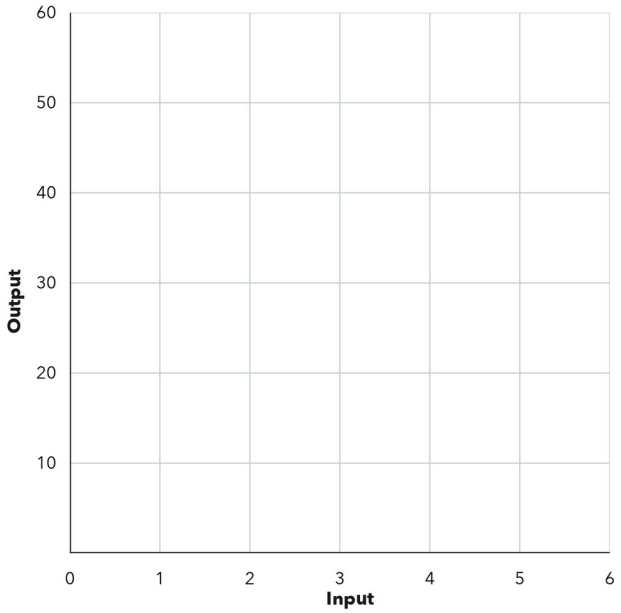
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Equations from Situations · 6.EE.C.9

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A sitter earns $8 per visit plus a one-time $5 travel payment. Define variables, write an equation, and connect each term to the story.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: multiplies the fixed fee by the number of items or leaves variables undefined.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?